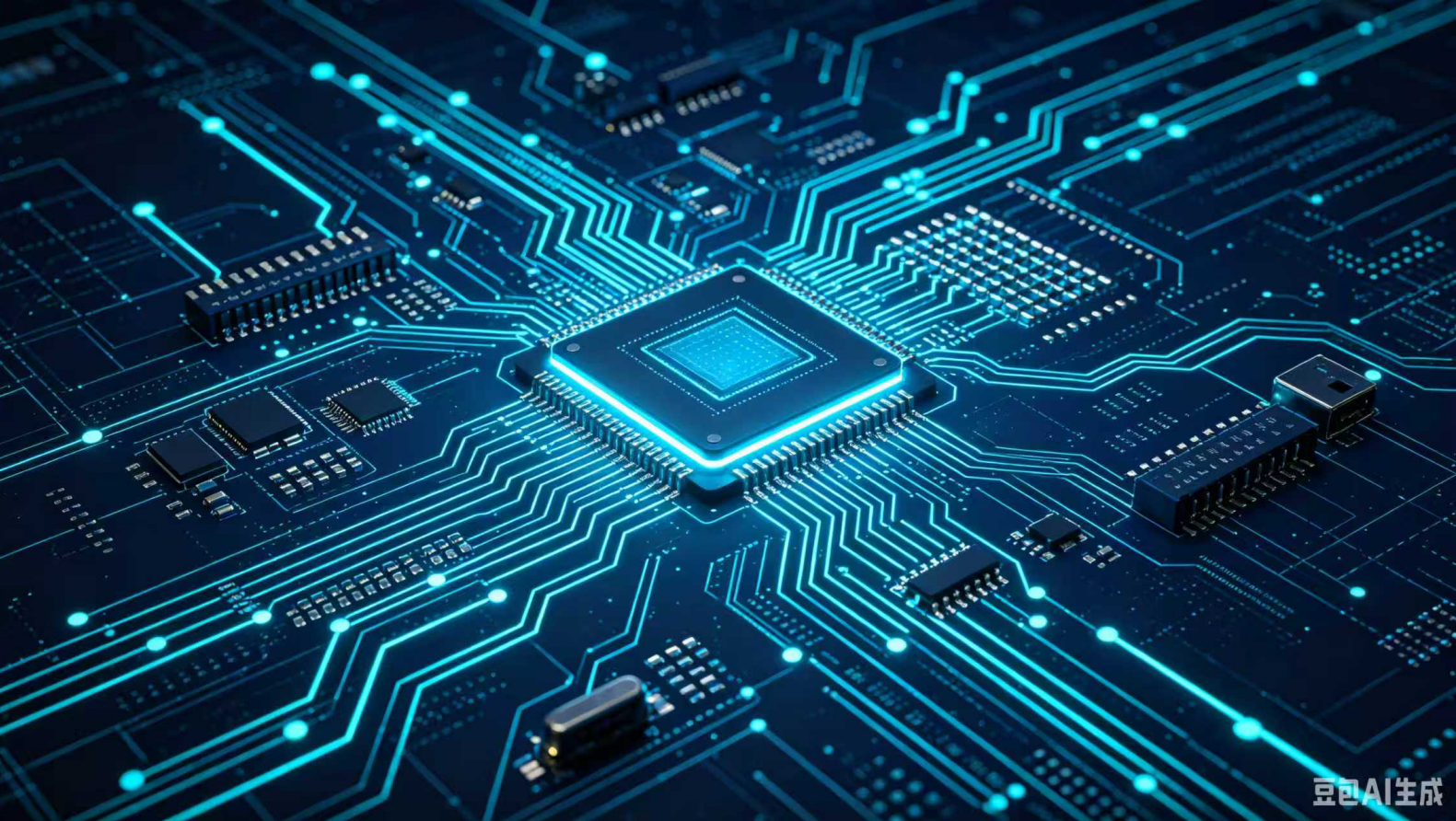




7. DC Circuits

Die Philosophen haben die Welt nur verschieden interpretiert; es kommt aber darauf an, sie zu verändern.

(English translation): Philosophers have hitherto only interpreted the world in various ways; the point is to change it.

– *Karl Marx, Theses on Feuerbach*, XI (written in 1845; the wording of Engels's edition of 1888; translation by the authors)

7.1 Concepts in Electric Circuits

An **electric circuit** is a closed path through which charge can flow, delivering energy, for example to a resistor or a light bulb. In order to achieve this, there must be a potential difference between the ends of the resistor or light bulb. To produce a steady flow, there must be a so-called **electromotive force (emf) device**, which maintains a potential difference between its two terminals and supplies the energy. A typical example of the emf device is a **battery**, which drives electron motion via electrochemical reactions.

The emf is represented by an arrow with a circle on its tail, pointing from the negative terminal to the positive one. When the source delivers energy to the circuit, the current inside it runs along the arrow, from the low-potential terminal to the high-potential one, and returns to the low-potential

terminal through the external circuit. When another source drives current through it backward, as when a battery is being charged, the current inside it runs against the arrow, and the source takes in energy.

We define the **emf** $\mathcal{E}$ of the emf device as the work per unit charge that the device does in moving charge from its low-potential terminal to the high-potential one:

$$\mathcal{E} = \frac{W}{q}$$

In order to study such electric circuits, we need also to know the current passing through a device, or the potential difference between its two ends. An **ammeter** is used to measure the current. To measure the current through a wire, the ammeter must be inserted in series: cut the wire and connect the two free ends to its terminals. Ideally, the ammeter should have no resistance so that it won't change the current it measures. In reality, the resistance of the ammeter is small, but not zero.

Similarly, to measure the potential difference, a **voltmeter** is used. It has to be connected to the two ends where the potential difference is to be measured. Ideally, the resistance of a voltmeter should be infinite so it won't change the potential difference it measures. In reality, it is large but is also finite.

Grounding a circuit means adopting the potential of the Earth — a conductor large enough to absorb or supply charge with no measurable change — as the zero reference for the circuit. In reality, it is done by connecting a specific position (grounding point) of the circuit to conducting moist dirt, rock or underground water below Earth's surface.

7.2 Energy Conservation in an Electric Circuit

Let's consider a simple circuit with an ideal emf device with constant emf $\mathcal{E}$ and a resistor R . A steady current i will be produced in the circuit in this case. Within a time interval dt , the amount of charge that moves through the emf device is therefore $dq = i dt$. Thus, the work done by the emf is $dU = \mathcal{E} dq = i \mathcal{E} dt$. The energy is dissipated by the resistor, and the rate of energy dissipation is:

$$P = iV_R = i^2 R = \frac{V_R^2}{R}$$

where $V_R = iR$ is the potential difference across the resistor. It equals $\mathcal{E}$ here only because this loop contains a single resistor. Due to the principle of energy conservation, the total rate of energy change is therefore:

$$\frac{dU}{dt} = i\mathcal{E} - i^2 R = 0$$

This is the simplest form of so-called **Kirchhoff's loop rule**: The algebraic sum of the potential changes encountered in a complete traversal of any loop of a circuit must be zero. To apply this rule to a loop in the circuit, we have to assign a direction of current for each part of the circuit, and then choose the direction of the loop (clockwise or counter-clockwise) to calculate. The potential changes are defined as:

1. For a resistor, if the current direction is the same as the loop direction, the potential change is $-iR$; otherwise it is $+iR$.
2. For an ideal emf device, if the emf arrow is along the loop direction, the potential change is $+\mathcal{E}$; otherwise it is $-\mathcal{E}$.

In addition to this, if we apply charge conservation to a surface surrounding a node in the circuit,

$$\oint_S \mathbf{j} \cdot d\mathbf{S} = 0$$

or equivalently, $\sum_{\alpha} i_{\alpha} = 0$ for all nodes. In this sum every current is counted as leaving the node.

Using the loop rules and charge conservation at each node, we can solve for the current in the circuit.

Example 7.1: Two resistors in series

Assuming that the two resistors R_1 and R_2 are connected to an emf of $\mathcal{E}$ (Figure 7.1), a steady current i flows through the emf. According to charge conservation, the current through both resistors is also i . Thus, the work done by the emf, and energy dissipated by the resistors become:

$$i\mathcal{E} = i^2 R_1 + i^2 R_2 \Rightarrow \mathcal{E} = i(R_1 + R_2)$$

This suggests that two resistors in series have equivalent resistance $R_{eq} = R_1 + R_2$ — a result already familiar.

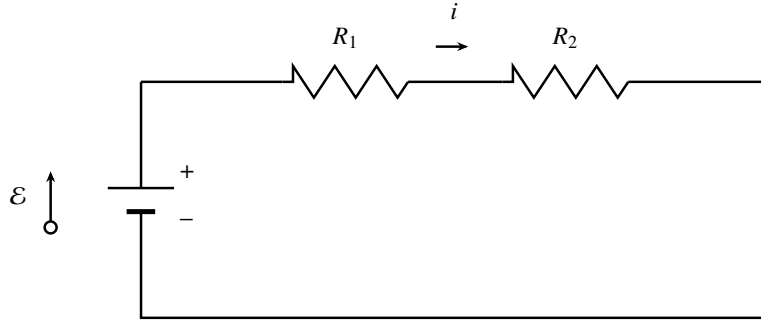


Figure 7.1: Two resistors R_1 and R_2 connected in series with an emf $\mathcal{E}$. The same current i flows through both, giving an equivalent resistance $R_{eq} = R_1 + R_2$.

Two resistors joined to the same two nodes are said to be **in parallel**. The loop rule applied to the loop they form gives the same potential difference V across both, and the node rule gives the total current $i = V/R_1 + V/R_2$. With $R_{eq} = V/i$,

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

Example 7.2: A cubic resistor network

A cubic network has resistance R on every edge (Figure 7.2). Find the equivalent resistance between two opposite corners.

We shall assume the opposite corners are P and P' . By symmetry, the currents in the three edges at P are equal, and so are those in the three edges at P' . Thus, if the total current entering at P is i , each of them carries $i/3$. By the symmetries of the cube that fix both P and P' — the three-fold rotations about the P – P' body diagonal together with the reflections in the planes through it — the six edges joining a neighbor of P to a neighbor of P' are all equivalent. At each neighbor of P the incoming current $i/3$ is shared by exactly two of them, so each of the six carries $i/6$. With these, the potential difference between P and P' is:

$$V = \frac{i}{3}R + \frac{i}{6}R + \frac{i}{3}R = \frac{5}{6}iR$$

Thus, the equivalent resistance between P and P' is $R_{eq} = 5R/6$.

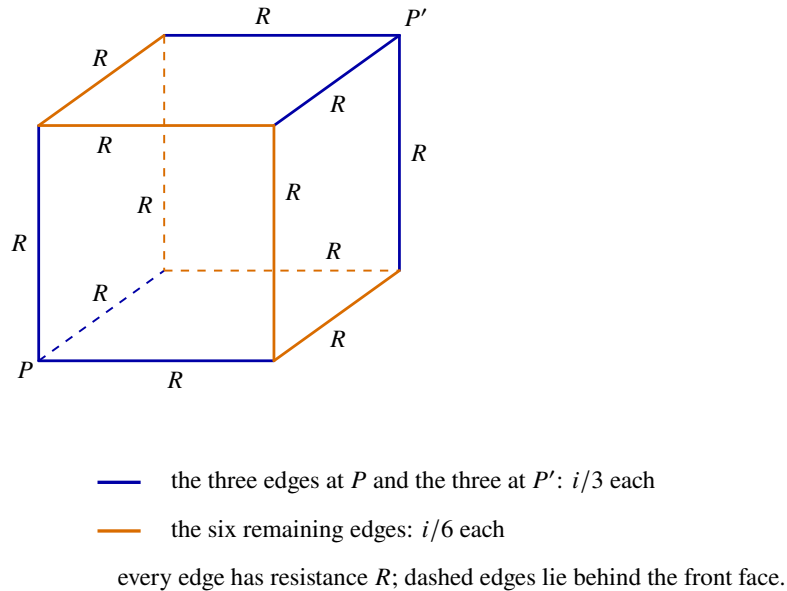


Figure 7.2: A cubic resistor network with each edge of resistance R . By symmetry, the three edges from P each carry $i/3$, the six middle edges each carry $i/6$, yielding an equivalent diagonal resistance $R_{eq} = 5R/6$.

You may wonder whether one can always deterministically solve for the current by applying the Kirchhoff loop rules and charge conservation. In fact, you can prove it using Euler's formula for a planar graph in graph theory.

7.2.1 Energy Stored in a Capacitor

Before we move on, we take a closer look at the capacitor. When we charge up a capacitor, we remove electrons from the positive plate and put them onto the negative plate. In the process, work must be done because the electric force is in the opposite direction. Suppose at a given instant, charge q' has already been transferred from one plate to another, both of which were initially charge

neutral. By definition, the potential difference at the moment is:

$$V' = \frac{q'}{C}$$

Thus, if an extra charge element dq' is being transferred, the work done is:

$$dW = V' dq' = \frac{q'}{C} dq'$$

Thus, the work done by transferring total charge of q can be obtained by integration:

$$W = \int_0^q \frac{q'}{C} dq' = \frac{q^2}{2C}$$

The work done is stored as potential energy U in the capacitor, therefore

$$U = \frac{q^2}{2C} = \frac{1}{2} CV^2$$

We can see that the energy can be expressed through the potential difference between the plates, which is directly related to the field between them. Assuming the capacitor is a large parallel-plate one, whose plates are separated by a distance d and have area S , its capacitance is the parallel-plate result $C = \epsilon_0 S/d$ of Chapter 6. Therefore, when the potential difference between its two plates is V , it stores potential energy:

$$U = \frac{1}{2} CV^2 = \frac{\epsilon_0 S}{2d} V^2$$

Ignoring fringing effects, the electric field inside is a uniform one with $E = V/d$. Therefore, the above expression can be written as:

$$U = \frac{1}{2} \epsilon_0 \left(\frac{V}{d} \right)^2 (Sd) = \frac{1}{2} \epsilon_0 E^2 (Sd)$$

Note Sd is nothing but the volume inside the capacitor, where the electric field exists. Thus, we can define the energy density of the electric field, u_E , by:

$$u_E = \frac{1}{2} \epsilon_0 E^2$$

Although we obtained the above expression for a very specific and special case, this expression holds for any electric field in vacuum. (Inside the linear dielectric of Section 6.4, the energy per unit volume is $\frac{1}{2} \epsilon E^2$ instead.) In fact, we can verify its validity in a few cases we are familiar with.

Example 7.3: Potential energy per length of a cylindrical capacitor.

We try two different methods (Figure 7.3). We shall assume that the inner and outer cylinders carry charges $+\lambda$ and $-\lambda$ per unit length.

Method 1: The capacitance is the coaxial result of Chapter 6, $C = 2\pi\epsilon_0 L / \ln\left(\frac{b}{a}\right)$, and $q = \lambda L$. Thus, the total energy per length is:

$$\frac{U}{L} = \frac{1}{L} \frac{q^2}{2C} = \frac{1}{L} \frac{\lambda^2 L^2}{4\pi\epsilon_0 L / \ln\left(\frac{b}{a}\right)} = \frac{1}{4\pi\epsilon_0} \lambda^2 \ln\left(\frac{b}{a}\right)$$

Method 2: The electric field is the line-charge field of Chapter 6, $E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}$. Therefore, the energy density of the electric field is:

$$u_E = \frac{1}{2} \epsilon_0 E^2 = \frac{1}{8\pi^2 \epsilon_0} \frac{\lambda^2}{r^2}$$

Thus, the total energy per length is:

$$\frac{U}{L} = \int_a^b 2\pi r dr \frac{1}{8\pi^2 \epsilon_0} \frac{\lambda^2}{r^2} = \frac{1}{4\pi\epsilon_0} \lambda^2 \ln\left(\frac{b}{a}\right)$$

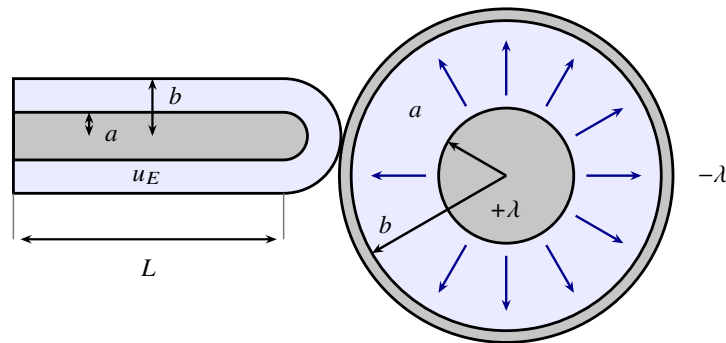


Figure 7.3: Energy stored in a cylindrical capacitor. The energy density $u_E = \frac{1}{2} \epsilon_0 E^2$ is integrated over the volume between the cylinders, yielding the same result as $U = q^2 / (2C)$.

Example 7.4: Potential energy of a spherical capacitor.

The inner and outer spheres have radii a and b and carry charges $+q$ and $-q$ (Figure 7.4). For two concentric spheres the field between them is that of the enclosed charge, $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$, so $V = \int_a^b E dr = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right) = \frac{q(b-a)}{4\pi\epsilon_0 ab}$, and $C = q/V = 4\pi\epsilon_0 \frac{ab}{b-a}$.

Method 1: With this capacitance, the total energy contained is:

$$U = \frac{q^2}{2C} = \frac{1}{8\pi\epsilon_0} q^2 \frac{b-a}{ab}$$

Method 2: With the field above, the energy density is:

$$u_E = \frac{1}{2} \epsilon_0 E^2 = \frac{1}{32\pi^2 \epsilon_0} \frac{q^2}{r^4}$$

The total energy is therefore:

$$U = \int u_E d^3\mathbf{r} = \int_a^b 4\pi r^2 \frac{1}{2} \epsilon_0 E^2 dr = \frac{1}{8\pi\epsilon_0} \int_a^b \frac{q^2}{r^2} dr = \frac{1}{8\pi\epsilon_0} q^2 \frac{b-a}{ab}$$

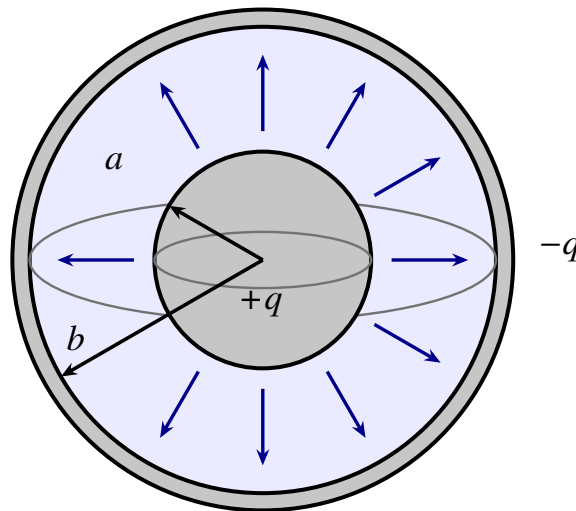


Figure 7.4: Energy stored in a spherical capacitor. Integrating the energy density u_E over the spherical shells between a and b yields $U = q^2(b-a)/(8\pi\epsilon_0 ab)$, consistent with $U = q^2/(2C)$.

7.3 RC Circuit

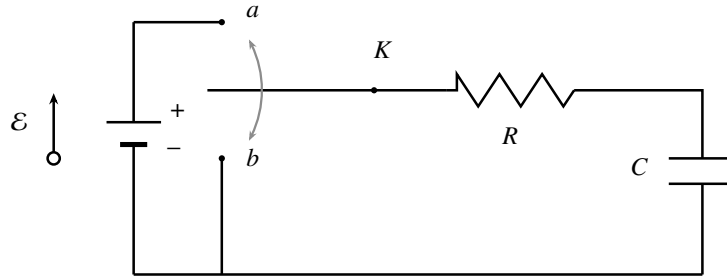


Figure 7.5: A typical RC series circuit consisting of a capacitor C , an ideal battery of emf $\mathcal{E}$, and a resistance R . When the switch K is thrown to a , the battery charges the capacitor; when thrown to b , the capacitor discharges through the resistor.

Figure 7.5 shows a typical RC circuit, which consists of a capacitor C and a series-connected resistor R . When the switch is thrown to a , the ideal battery $\mathcal{E}$ is connected into the circuit, and the battery charges the capacitor; the potential difference across the capacitor approaches $\mathcal{E}$. If the switch is thrown to b , the battery is not connected. Then if the capacitor is charged, it drives a current through the resistor and begins to discharge. The current goes through the resistor in both cases.

In the charging process, the work done by the emf is partially dissipated by the resistor, and the rest is stored on the capacitor. Assuming the capacitor has charge q at a specific moment, we have:

$$\begin{aligned} i\mathcal{E} &= i^2R + \frac{d}{dt} \left(\frac{q^2}{2C} \right) \\ &= i^2R + i \frac{q}{C} \end{aligned}$$

The relation $i = \frac{dq}{dt}$ is employed above. This immediately yields:

$$\frac{\mathcal{E}}{R} = \frac{dq}{dt} + \frac{q}{RC}$$

Since $\frac{q}{RC}$ must have the same dimension as $\frac{dq}{dt}$, which is a current, RC has the dimension of time; it is characteristic of this circuit. We define $\tau = RC$, and the solution with $q(0) = 0$ is:

$$q = C\mathcal{E}(1 - e^{-t/\tau})$$

The current is therefore:

$$i = \frac{dq}{dt} = i_0 e^{-t/\tau}$$

where $i_0 = \mathcal{E}/R$ is its largest value, at $t = 0$.

In the discharging process, the capacitor drives a current through the resistor, and thus the lost energy of the capacitor is dissipated by the resistor. Therefore

$$\frac{d}{dt} \left(\frac{q^2}{2C} \right) + i^2R = 0$$

Again, we employ $i = \frac{dq}{dt}$ and arrive at:

$$\frac{dq}{dt} + \frac{q}{RC} = 0$$

The solution with $q(0) = C\mathcal{E}$, the charge approached after a long charging, is:

$$q = C\mathcal{E}e^{-t/\tau}$$

and

$$i = -i_0 e^{-t/\tau}$$

7.4 Exercises

1. A metallic cylinder (radius a , length l) and a metallic cylindrical shell (radius b , length l) are coaxial, and the region between $a < r < b$ is a vacuum. The inner cylinder carries charge Q and the outer shell carries charge $-Q$. For simplicity, we ignore the boundary effects.
 - (a) What is the energy density u_E at radius r ($a < r < b$)?
 - (b) What is the total potential energy U of the system?
 - (c) What is the capacitance C of the system?
2. Consider a DC circuit composed of 4 resistors and an ideal emf as shown in the figure. The node D can be tuned freely along r_4 ; let x be the resistance of the part of r_4 between B and D, and assume $r_4 \geq |r_1 - r_3|$. Find the value of x for which the current i in the branch AD is smallest, and that smallest current.

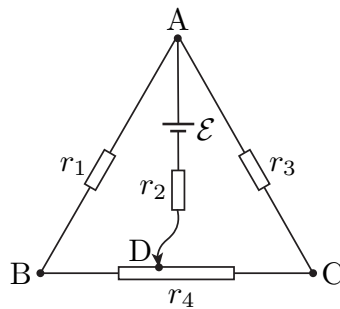


Figure 7.6: DC circuit for problem 2.

3. Consider the Wheatstone bridge as shown in the figure. Take I positive from C to D.
 - (a) Compute the current I going through the resistor r_g .
 - (b) What is the condition for I to be zero?

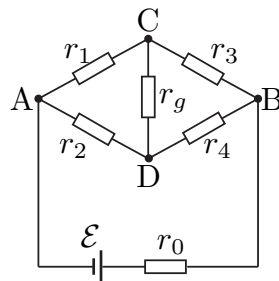


Figure 7.7: DC circuit for problem 3.

4. Consider the RC circuit as shown in the figure, with an ideal emf. The capacitor was not charged before the switch was closed at $t = 0$. Let Q be the magnitude of the charge on either plate. What is the time dependence of Q at time $t > 0$?

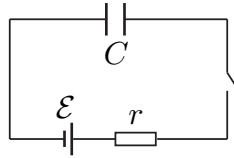


Figure 7.8: RC circuit for problem 4.

5. Consider the RC circuit shown in the figure, with an ideal emf. The capacitor is uncharged when the switch is closed at $t = 0$. Let Q be the magnitude of the charge on either plate. Find the time dependence of Q for $t > 0$, and its limit as $t \rightarrow \infty$.

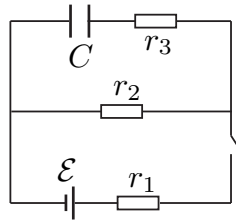


Figure 7.9: RC circuit for problem 5.

6. An unknown resistor R is measured with an ammeter of resistance R_A and a voltmeter of resistance R_V , powered by a battery modeled as an ideal emf $\mathcal{E}$ in series with a resistor r . All resistances are positive. The ammeter reads the current i through it, the voltmeter reads the potential difference V between its terminals, and the measured resistance is taken to be $R_m = V/i$. In both of the following connections the battery, the ammeter and R are joined in a single loop. In connection (i) the voltmeter's terminals are connected to the two ends of R . In connection (ii) they are connected to the end of the ammeter that is not joined to R and to the end of R that is not joined to the ammeter.
- Find R_m for each connection. Does it depend on $\mathcal{E}$ or r ?
 - Find the relative error $(R_m - R)/R$ for each connection, and the value R^* of R at which the two errors have equal magnitudes. Which connection gives the smaller error when $R > R^*$?
7. An ideal emf device of emf $\mathcal{E} > 0$ is connected between nodes A and B , with its emf arrow pointing from A to B . Two branches connect B back to A : the first contains a resistor R_1 , a node C and a resistor R_2 , in that order from B to A ; the second contains only a resistor R_3 . All wires are ideal, $R_1, R_2, R_3 > 0$, and the currents are steady. Let i_{12} be the current from B through R_1 , C and R_2 to A , let i_3 be the current from B through R_3 to A , and let i be the current through the emf device from A to B .
- Use the loop and node rules to find i_{12} , i_3 and i , and the equivalent resistance $R_{eq} = \mathcal{E}/i$ seen by the emf device.
 - Node A is grounded; a single ground connection carries no current. Find the potentials V_A , V_B and V_C . The ground connection is then moved from A to C . Find the three potentials again. Which of the quantities found in (a) change?
 - Show that the power supplied by the emf device equals the total power dissipated in the three resistors.

8. Two batteries are modeled as ideal emf devices $\mathcal{E}_1$ and $\mathcal{E}_2$ in series with resistors r_1 and r_2 , where $\mathcal{E}_1$, $\mathcal{E}_2$, r_1 and r_2 are all positive. The positive terminals of the two batteries are joined to one end of a load resistor $R > 0$, and their negative terminals to its other end. All wires are ideal, and the currents are steady.
- Find the current i_R through R , taken positive from the end joined to the positive terminals toward the other end, and the current i_2 in battery 2, taken positive when it leaves battery 2 at its positive terminal.
 - Find the condition on $\mathcal{E}_2$ under which battery 2 is being charged, that is, $i_2 < 0$. What does the condition become when the load is removed ($R \rightarrow \infty$), and when $\mathcal{E}_1 = \mathcal{E}_2$?
9. A parallel-plate capacitor in vacuum has plates of area S . The lower plate is fixed, and the upper plate can be moved perpendicular to the plates; x is the separation between them. Neglect edge effects. The upper plate is moved slowly, so that its kinetic energy and any heat produced can be neglected. Let F be the electric force on the upper plate, taken positive in the direction of increasing x .
- The plates are isolated and carry charges $+q$ and $-q$. Find the stored energy $U(x)$, and from the change of U in a small displacement dx , find F .
 - The plates are instead connected to an ideal battery that keeps their potential difference at V . Find $U(x)$, and the work dW_b done by the battery when x increases by dx . Explain why $F = -dU/dx$ now gives the wrong sign, find F from energy conservation, and compare with (a).
10. Two ideal capacitors have capacitances $C_1 > 0$ and $C_2 > 0$. Their lower plates are joined by an ideal wire, and this common node is taken as the zero of potential. At first their upper plates are isolated, at potentials V_1 and V_2 , where $V_1 > V_2 \geq 0$, so that they carry charges $C_1 V_1$ and $C_2 V_2$. At $t = 0$ an ideal switch is closed that joins the two upper plates through a resistor $R > 0$. There is no emf device in the circuit, and radiation is neglected. Let $q_1(t)$ and $q_2(t)$ be the charges on the upper plates, and $i(t)$ the current through R , positive from the upper plate of C_1 toward that of C_2 .
- Using charge conservation, find the final common potential V_f of the upper plates and the final charges q_1 and q_2 .
 - Use charge conservation and the loop rule to find $i(t)$, $q_1(t)$ and $q_2(t)$ for $t \geq 0$, and the time constant τ .
 - Find the initial stored energy U_i , the final stored energy U_f , and the heat $H = \int_0^\infty i^2 R dt$ produced in the resistor. Show that $H = U_i - U_f$, and that H does not depend on R .